

Required System Configuration

Minimum viable specification for every environment. All values below are mandatory, not advisory.

1. Developer workstation

Operating system	macOS 12+, Windows 10/11 with WSL2, or Ubuntu 22.04+
CPU	2 cores
RAM	8 GB
Free disk	10 GB
Node.js	18.x LTS
Package manager	pnpm 8+
Go	1.21+
Git	2.40+
Browser	Chromium, Chrome or Edge, current
Network	Outbound HTTPS to source control and the PostgreSQL host
Credentials	Non-production DSN and non-production admin account

2. Application host (staging or production)

Operating system	Ubuntu 22.04 LTS
CPU	2 vCPU
RAM	4 GB
Disk	40 GB SSD
Persistent volume	Durable mount for backend/uploads/
Node.js	18.x LTS
Go toolchain	1.21+ (build only)
Process manager	PM2, with pm2 startup and pm2 save
Reverse proxy	Nginx or Caddy terminating TLS
Frontend port	3000 (peepal-frontend)
API port	8080 (PORT ; .env.example uses 3001 locally)
Public ports	443 only; 80 redirecting to 443
DNS	Wildcard record for tenant subdomains, plus an API hostname
Outbound	HTTPS to the database and the SMTP relay

3. Database

Engine	PostgreSQL 14+
Driver	GORM with pgx
Storage	10 GB
Transport	TLS enforced
Access control	IP allowlist or private networking
Backups	Daily automated snapshots
Schema changes	Only via main.go --migrate, run deliberately

4. Required environment variables

Variable	Requirement
DB_PATH	PostgreSQL DSN
PORT	API listen port; falls back to 8080
COOKIE_DOMAIN	Cookie scope. Must be empty on localhost
JWT / admin seed secrets	Unique per environment
SMTP settings	Required for invitations and notifications
NEXT_PUBLIC_API_URL	Origin of the Go API
NODE_ENV / PORT	production / 3000 on the host

Never commit a real .env or .env.local. Inject secrets at the host or through a secret manager.

5. End-user device

Browser	Chrome, Edge, Firefox or Safari, within the last two major versions
Screen	1280 × 720
Cookies	First-party cookies allowed for the Peepal domain
JavaScript	Enabled
Network	Stable broadband, no captive portal

Desktop required for dense admin tables and bulk CSV tools. Mobile browsers are supported for everyday modules only.

6. Non-negotiable operating rules

- Sessions use the httpOnly cookie peepal_token, valid 24 hours. Cookies must not be stripped by the proxy.
- The proxy must forward the tenant header and cookies unchanged.
- backend/uploads/ must be included in the backup policy alongside the database.
- Change the seeded administrator credentials before the instance is reachable by anyone else.
- Login is blocked without an active subscription on the tenant.